

Numerical simulations of PBH formation via delayed FOPTs

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Outline

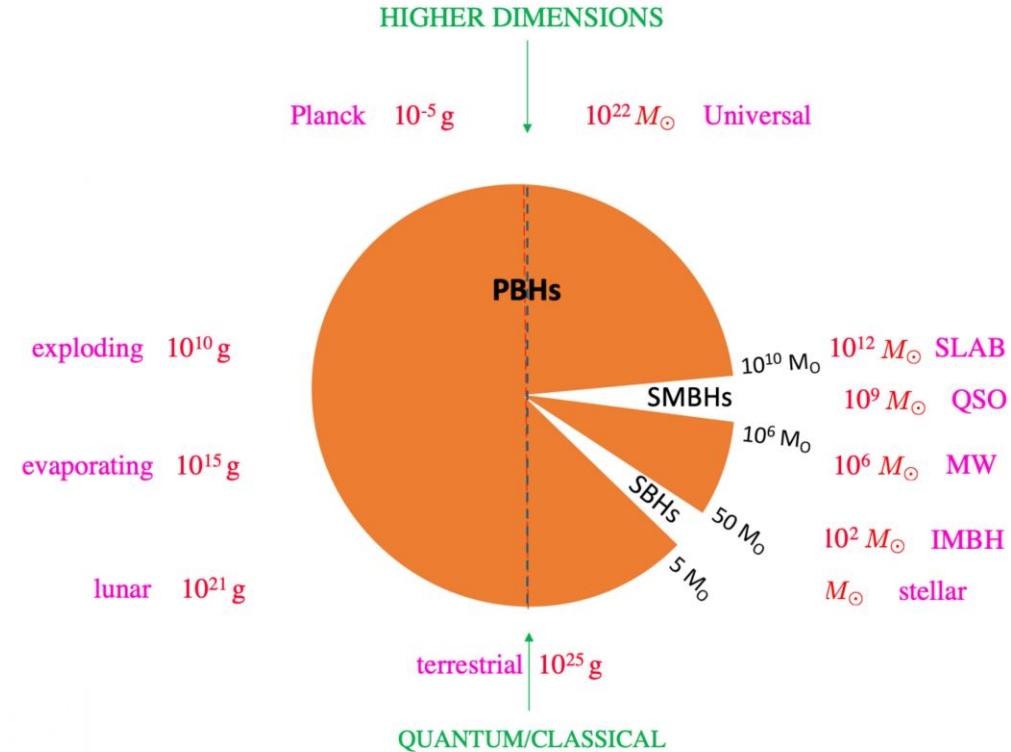
- Introduction
- Numerical simulation setups
- Dynamics of false vacuum domains
- Thresholds for type A/B PBHs
- Conclusions and discussions

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Primordial black holes (PBHs)

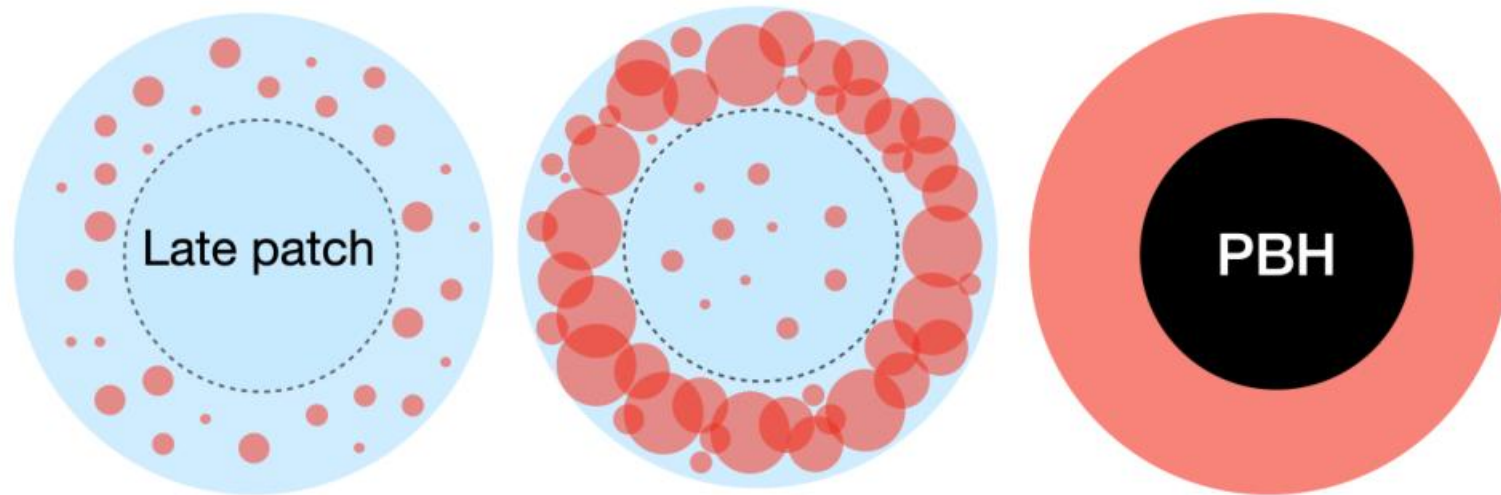
- Why PBHs?
 - Unique probe of small-scale primordial perturbations
 - Candidate for Cold Dark Matter
 - Progenitors of LIGO-Virgo-KAGRA binary mergers
 - Seeds for supermassive black holes
- Formation mechanisms:
 - Collapse of large adiabatic/isocurvature perturbations
 - (P)reheating dynamics
 - Collapse of topological defects (cosmic strings, domain walls)
 - **First-order phase transitions (FOPTs)**



[Carr, Green 2406.05736]

PBHs from delayed FOPTs

- Bubble nucleation is stochastic
- Strongly supercooled phase transitions tend to be **delayed**
- Some regions remain in the **false vacuum domains (FVDs)**
 - Outside FVD: Radiation dominated (dilutes as $\rho_R \sim a^{-4}$)
 - Inside FVD: Vacuum energy dominated (**constant** ρ_V)



- Result: A **large overdensity** develops inside the FVD, potentially collapsing into a PBH

PBH formation conditions



➤ **Key question:** Under what conditions does an FVD collapse into a PBH?

➤ Traditional **critical overdensity criterion**:

- $\delta = \frac{\rho_{in} - \rho_{out}}{\rho_{out}} > \delta_c \sim 0.45$
- Adopted from curvature perturbation collapse

➤ Recent criticisms:

- [Flores, Kusenko, Sasaki PRD2024]: Only expanding solutions exist; need **Schwarzschild criterion** $2M/R > 1$
- [Franciolini et al. 2503.01962; Wang et al. 2601.14412]: Gauge dependence issue

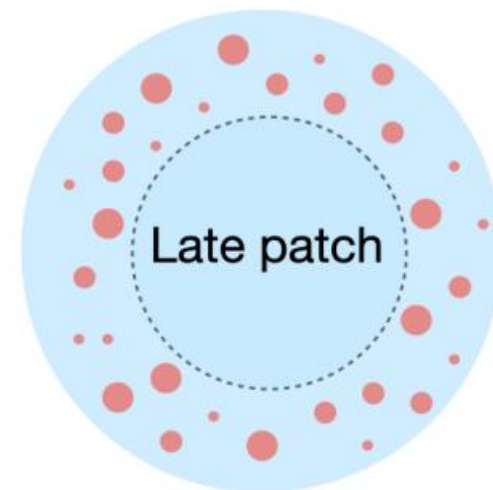
Dynamics of FVDs

➤ Key assumptions:

- FVD is **spherically symmetric**
- **No new nucleation** in the FVD

➤ Dynamical outcomes of FVDs:

- Supercritical PBH with an interior baby universe
 - type B PBH with a **bifurcating trapping horizon** ($\Theta^+ = \Theta^- = 0$)
- Subcritical (normal) PBH
 - type A PBH with an **apparent horizon** ($\Theta^+ = 0, \Theta^- < 0$)
- Vanish as phase transition completes?

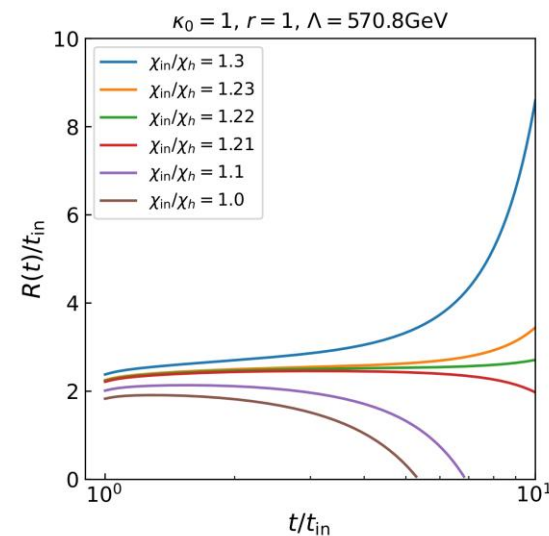


Dynamics of FVDs

➤ Previous work:

- Thin-wall approximation
- Israel's junction conditions to match interior and exterior metrics

[Hashino et al. JCAP2025]



$$\ddot{\chi} + (4 - 3a^2\dot{\chi}^2)H\dot{\chi} + \frac{2}{a^2\chi}(1 - a^2\dot{\chi}^2) = \left(\frac{3\sigma}{4M_{\text{pl}}^2} - \frac{\rho_V}{\sigma} \right) \frac{(1 - a^2\dot{\chi}^2)^{3/2}}{a}$$

➤ Two kinds of solutions:

- FVD boundary radius tends to infinity --- supercritical PBH
- FVD boundary radius reaches zero --- **cannot distinguish subcritical PBH or dispersal!**

Our goal: Perform **fully nonlinear numerical simulations** of coupled gravity-scalar-fluid systems to study the dynamics and determine robust formation criteria

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The model

- Coupled gravity-scalar-fluid system

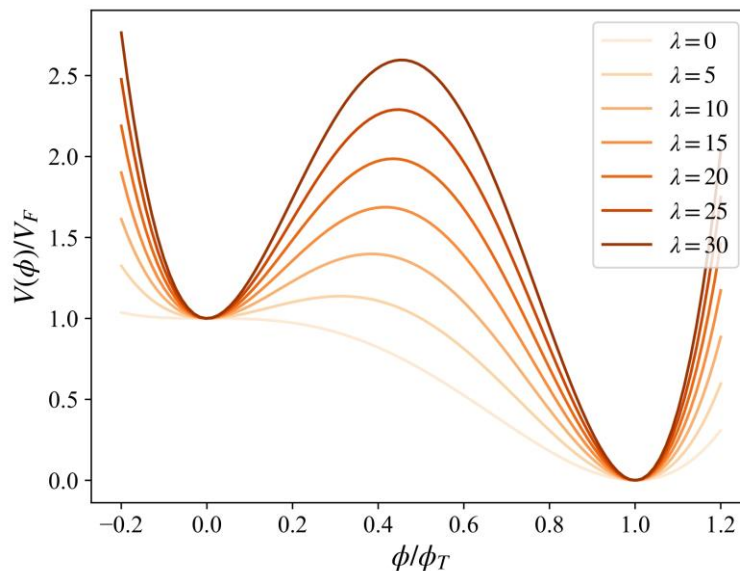
$$S = \int d^4x \sqrt{-g} \left[\frac{{}^{(4)}R}{16\pi} - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) + \mathcal{L}_{\text{fluid}} \right]$$

- Spherically symmetric metric (geodesic slicing $A = 1$)

$$ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega^2$$

- Scalar field: phenomenological potential $V(\phi) = \left[1 + \lambda \left(\frac{\phi}{\phi_T} \right)^2 - (2\lambda + 4) \left(\frac{\phi}{\phi_T} \right)^3 + (\lambda + 3) \left(\frac{\phi}{\phi_T} \right)^4 \right] (V_F - V_T) + V_T$

- True vacuum: $\phi = \phi_T, V_T = 0$
- False vacuum: $\phi = 0, \rho_V = V_F$
- $\lambda \rightarrow \infty$: Thin-wall limit
- $\lambda \rightarrow 0$: Thick-wall limit



- Fluid: radiation fluid ($p = \rho/3$), interacting with scalar only via gravity

Initial conditions & Numerical implementation

- Superhorizon FVD embedded in radiation FLRW background

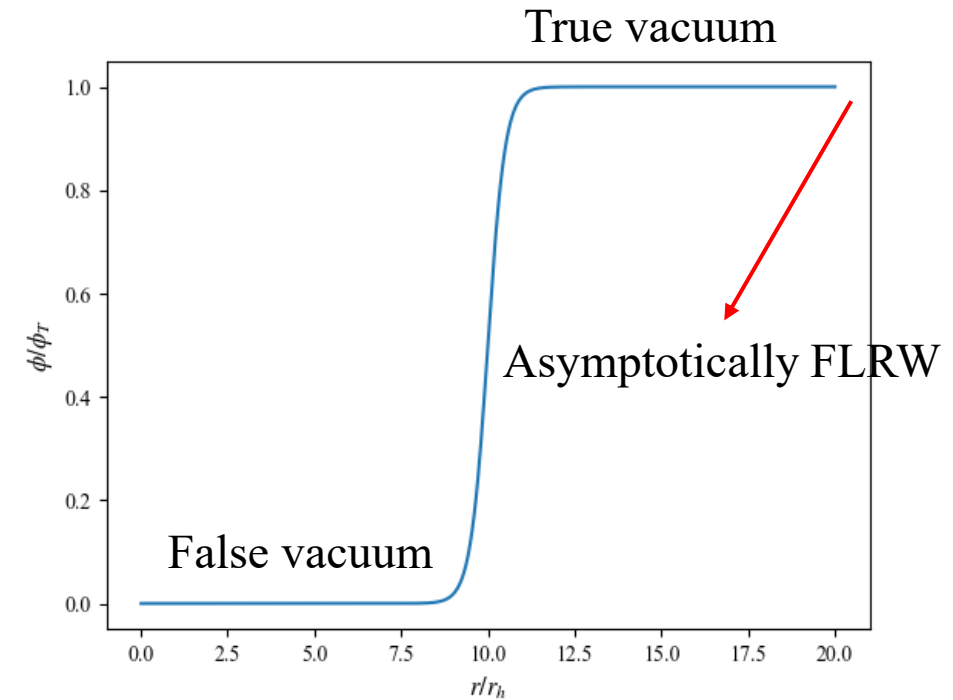
$$\phi(t_i, r) = \frac{\phi_T}{2} \left[1 + \tanh \left(\frac{r - r_i}{l} \right) \right]$$

- Initial wall thickness $l = \frac{r_i}{10}$
- To satisfy Hamiltonian constraint

$$\rho(t_i, r) = \rho_b(t_i) - \rho_\phi(t_i, r)$$

- Numerical Implementation

- Finite-difference spatial discretization
- Runge-Kutta time integration
- Adaptive mesh refinement to resolve bubble wall



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Three dynamical outcomes

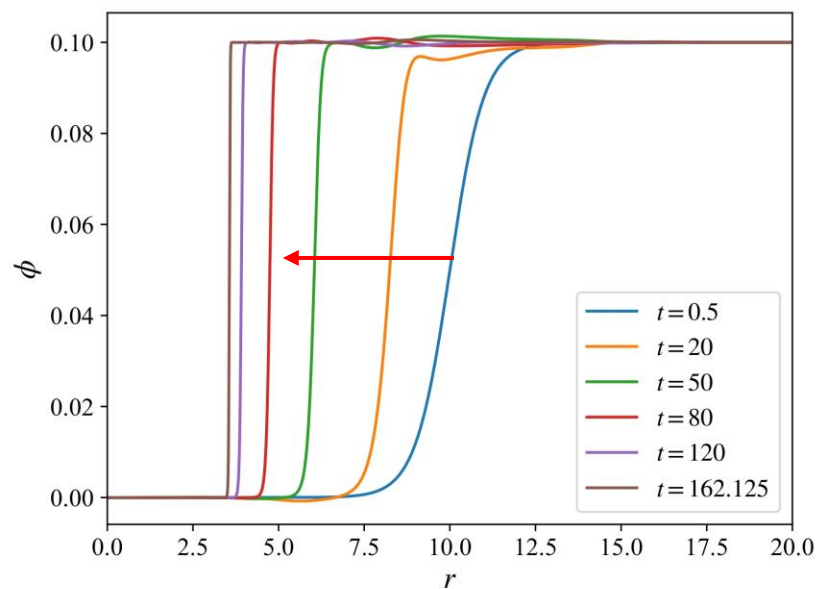
- Representative parameter set

$$\phi_T = 0.1, \quad \lambda = 20, \quad r_i = 10$$

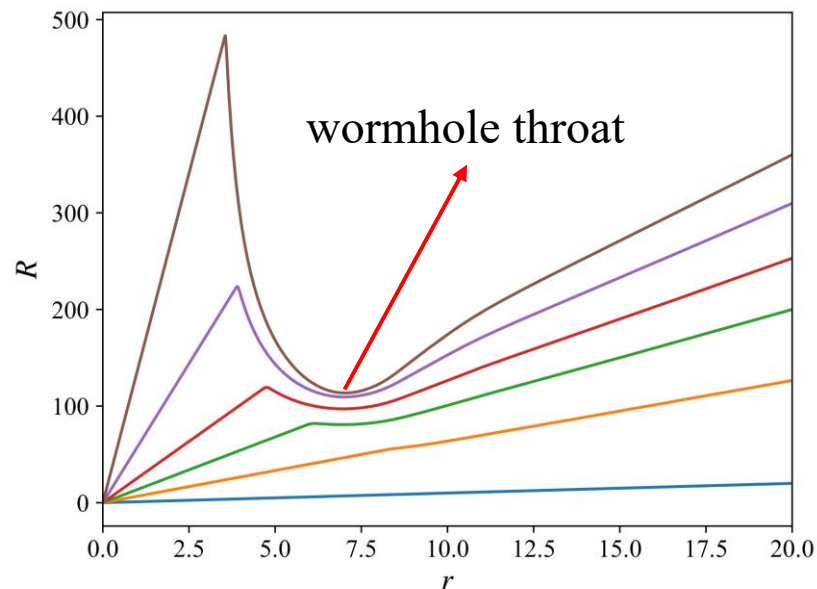
- Depending on the vacuum energy density V_F , we identify three distinct fates for the FVD:
 - Type B PBH (Supercritical): Baby universe formation + wormhole pinch-off
 - Type A PBH (Subcritical): Direct collapse of the bubble wall
 - No PBH: FVD dispersal and scalar wave radiation

Outcome 1: Type B PBH (Supercritical)

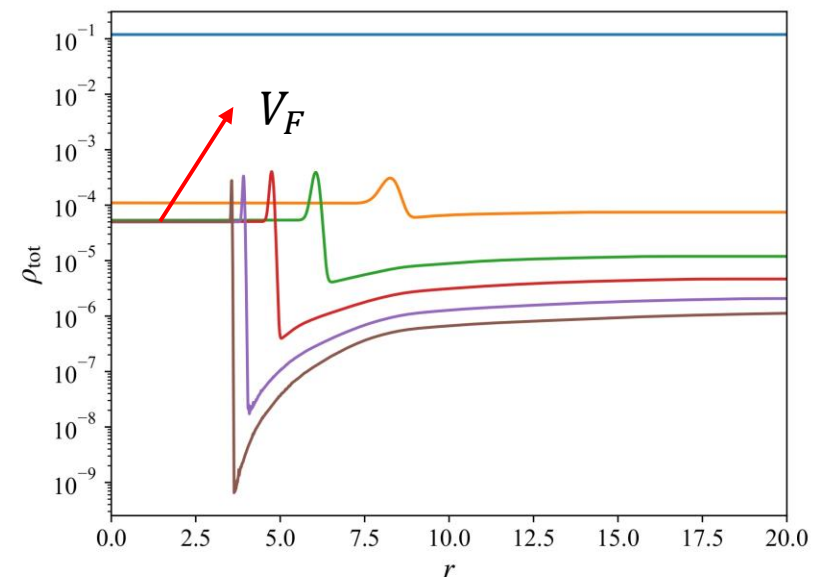
- $V_F = 5 \times 10^{-5} H_i^2$, vacuum energy dominates
- Vacuum energy domination before horizon crossing



FVD shrink



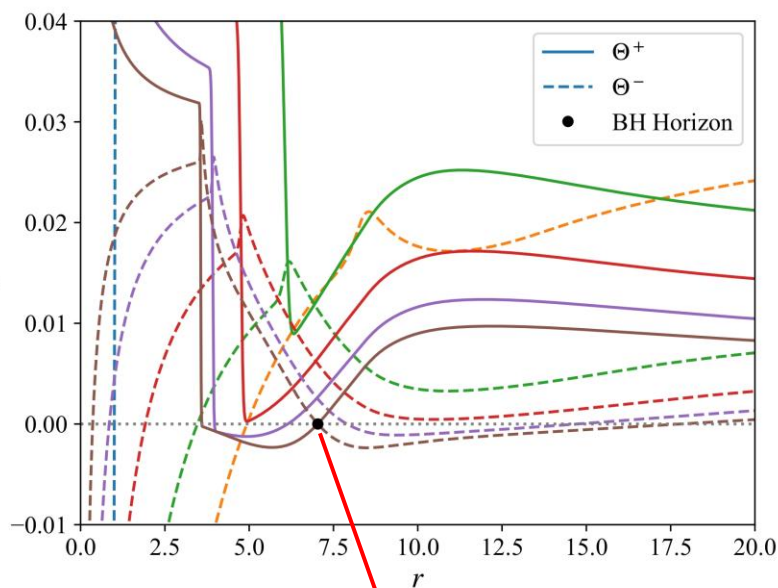
exponential expansion inside FVD



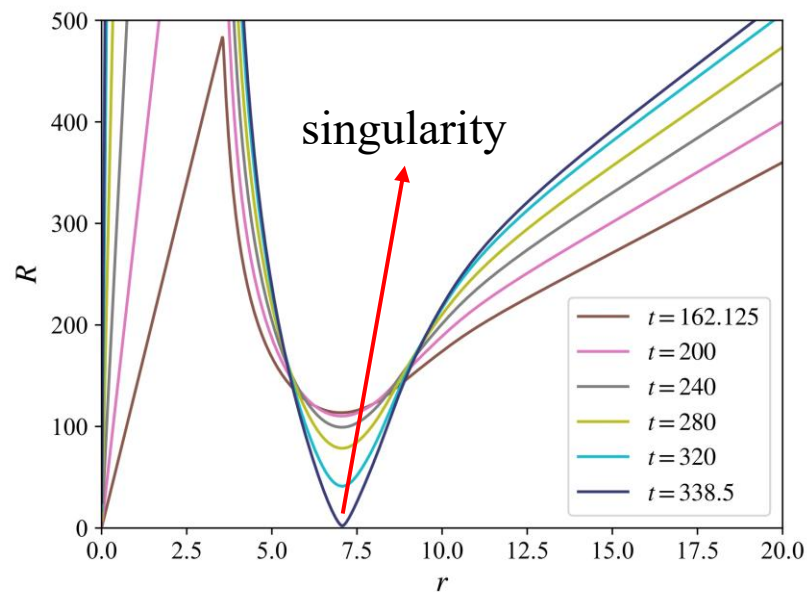
large density contrast

Outcome 1: Type B PBH (Supercritical)

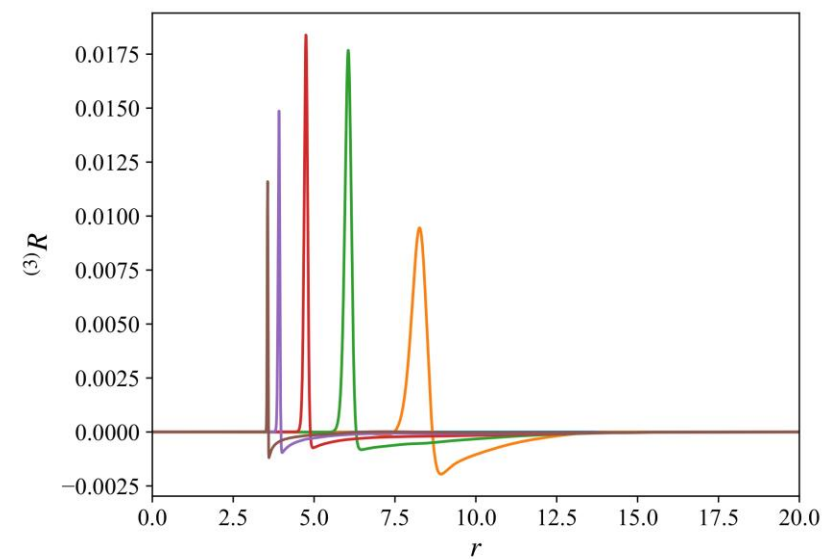
- $V_F = 5 \times 10^{-5} H_i^2$, vacuum energy dominates
- Vacuum energy domination before horizon crossing



bifurcating trapping horizon



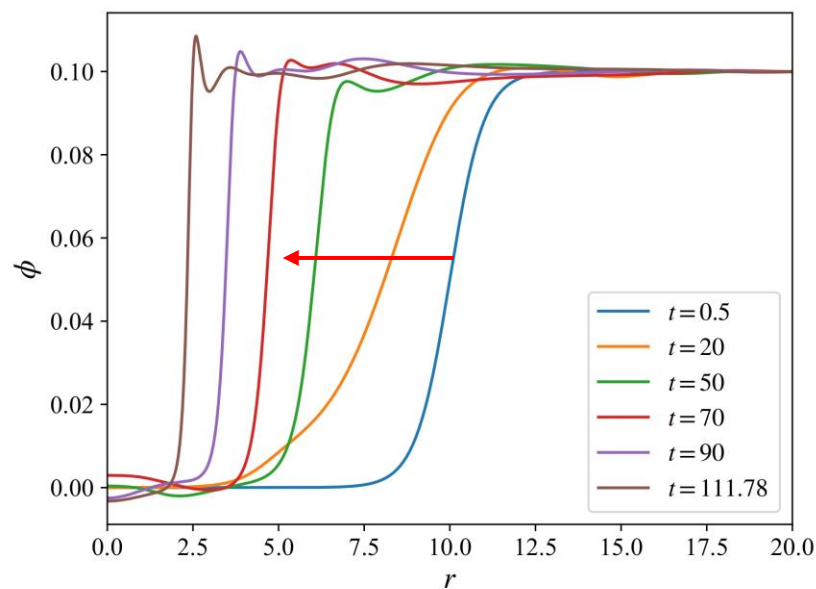
wormhole pinches off



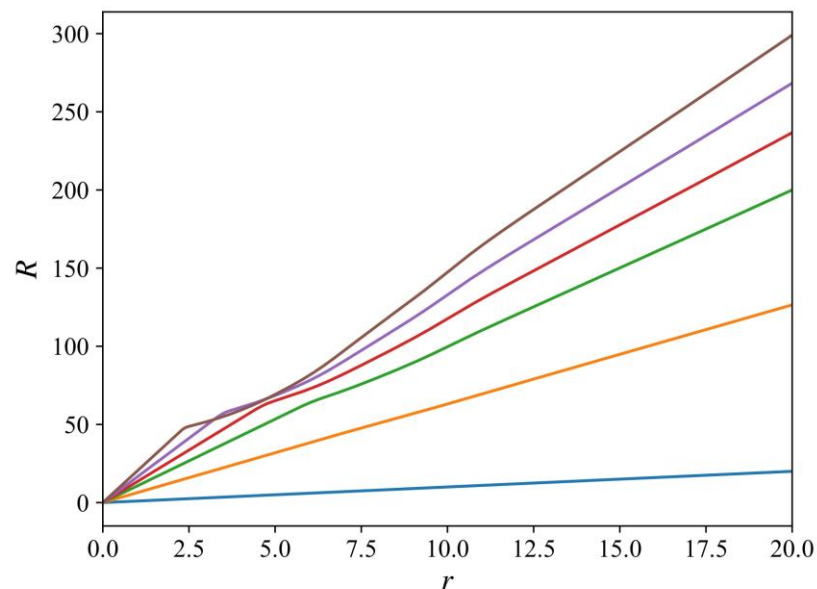
local spatial curvature

Outcome 2: Type A PBH (Subcritical)

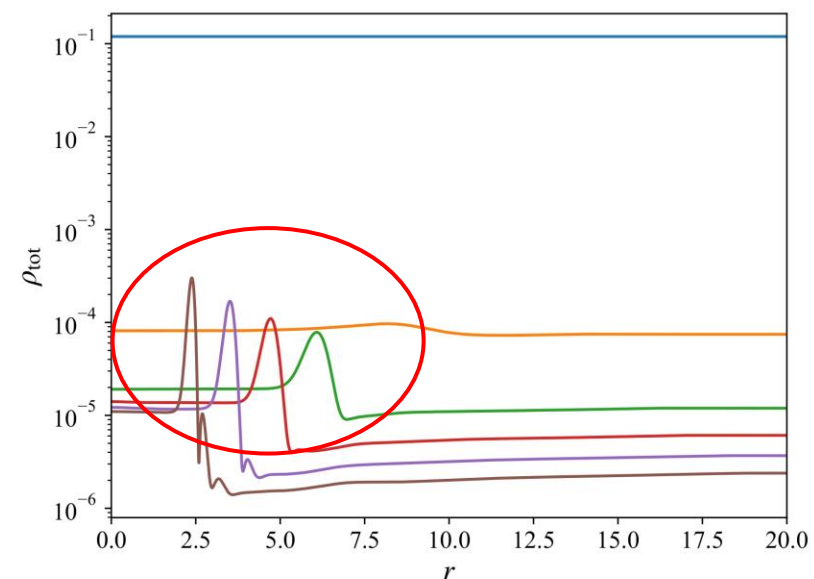
- $V_F = 1 \times 10^{-5} H_i^2$, bubble wall collapses
- Verify analytical analysis of [\[Flores, Kusenko, Sasaki PRD2024\]](#)



FVD shrink



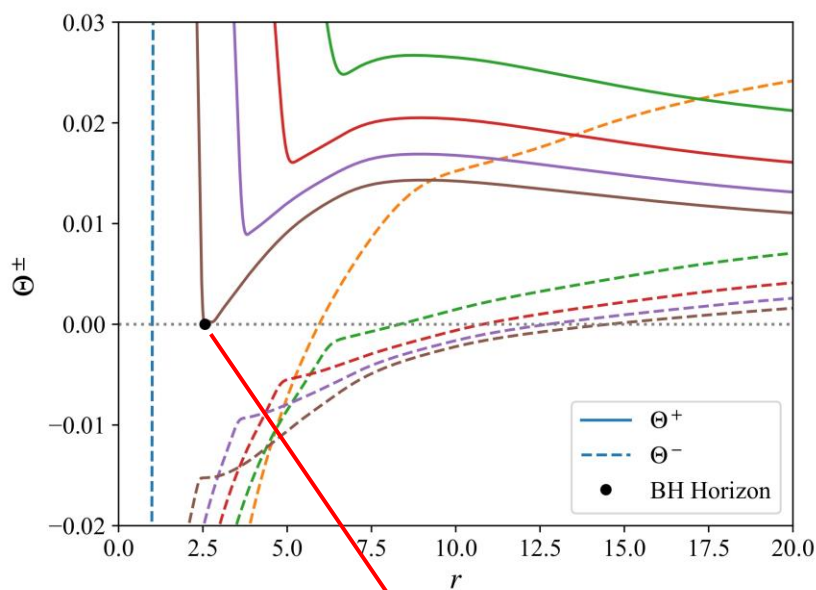
monotonical



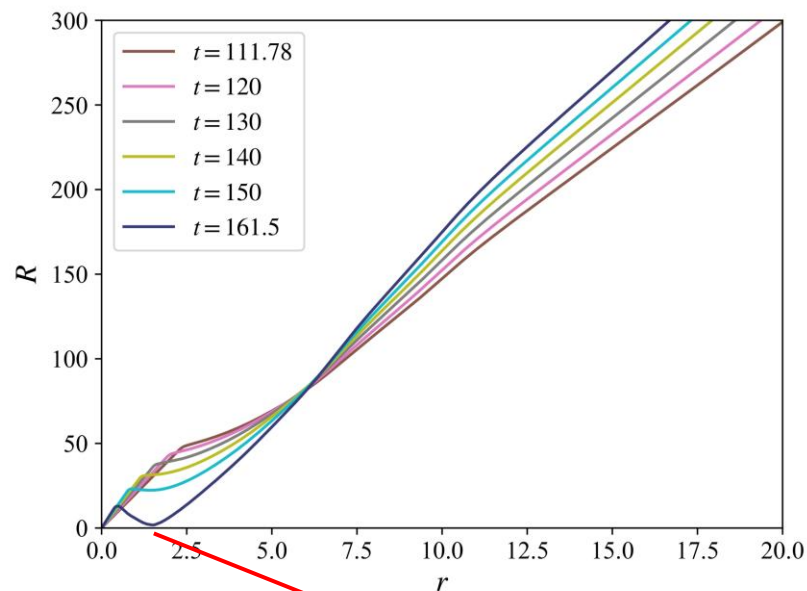
wall energy concentration

Outcome 2: Type A PBH (Subcritical)

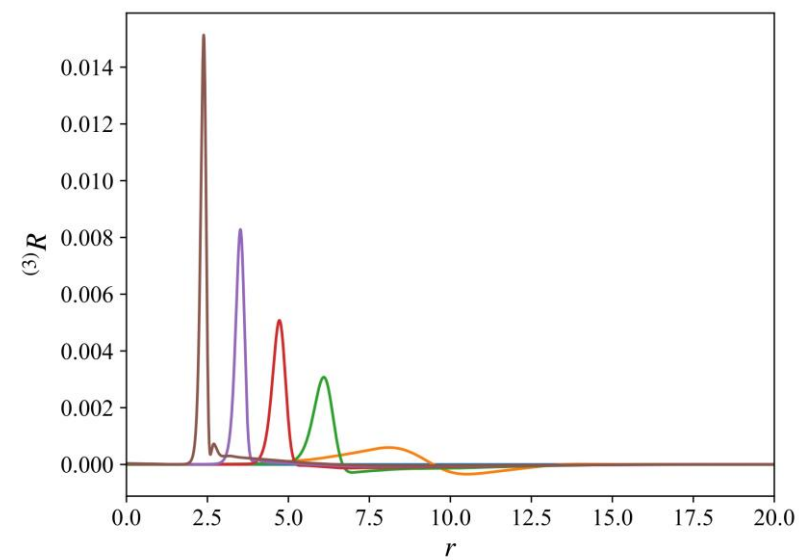
- $V_F = 1 \times 10^{-5} H_i^2$, bubble wall collapses
- Verify analytical analysis of [\[Flores, Kusenko, Sasaki PRD2024\]](#)



apparent horizon



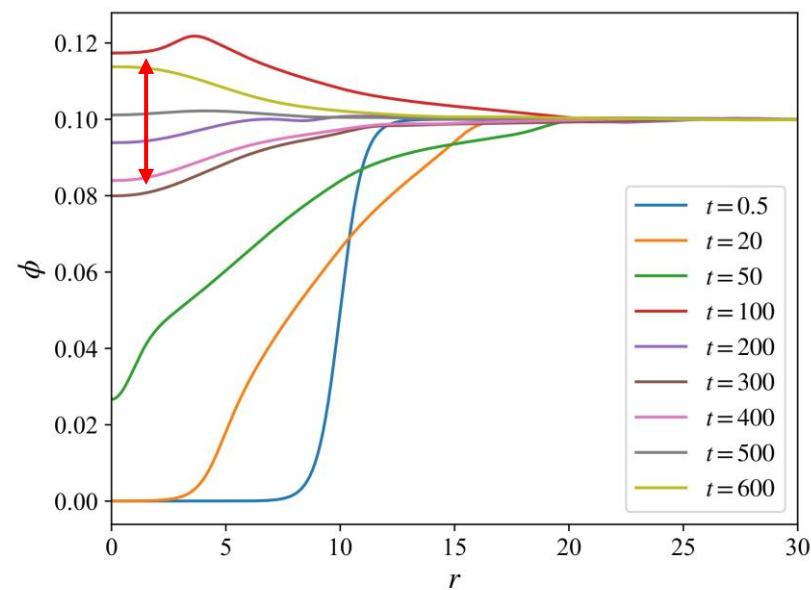
singularity



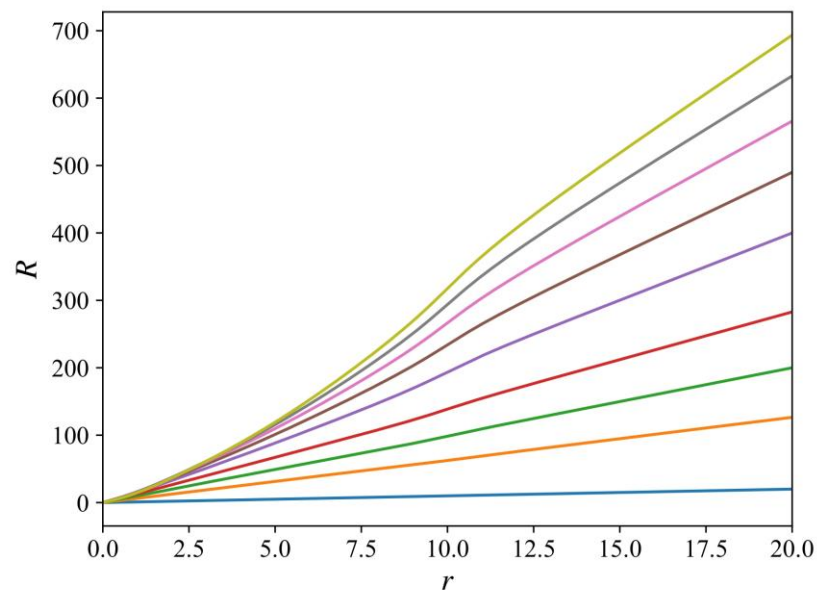
local spatial curvature

Outcome 3: No PBH Formation (Dispersal)

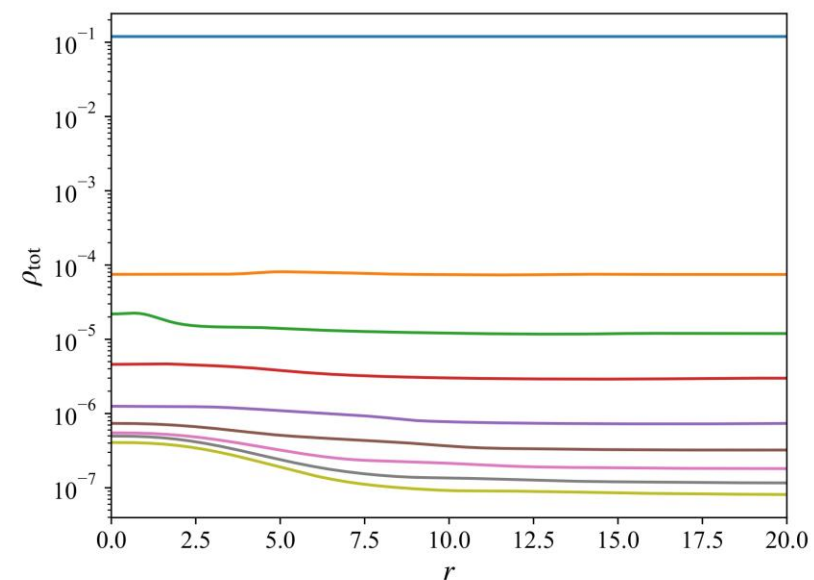
- $V_F = 5 \times 10^{-7} H_i^2$, no PBH formed
- Energy is radiated away as outgoing scalar waves



oscillate about true vacuum



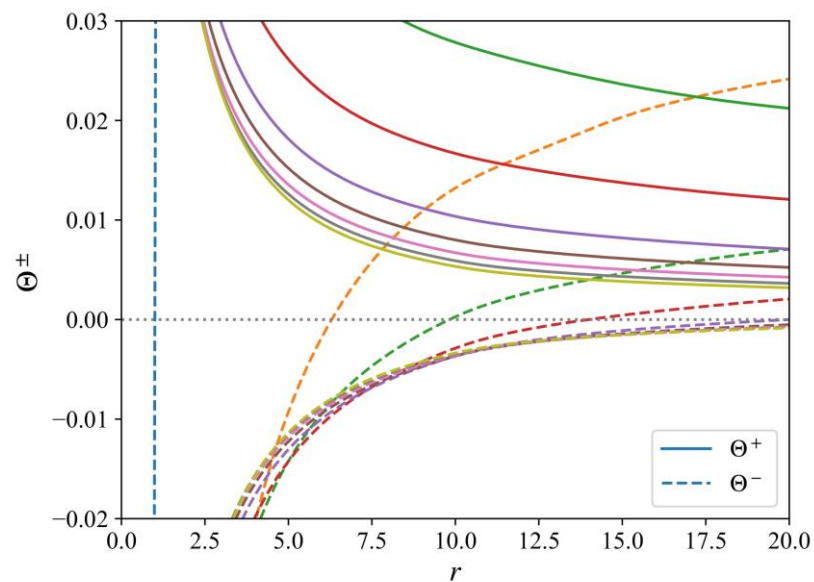
monotonical



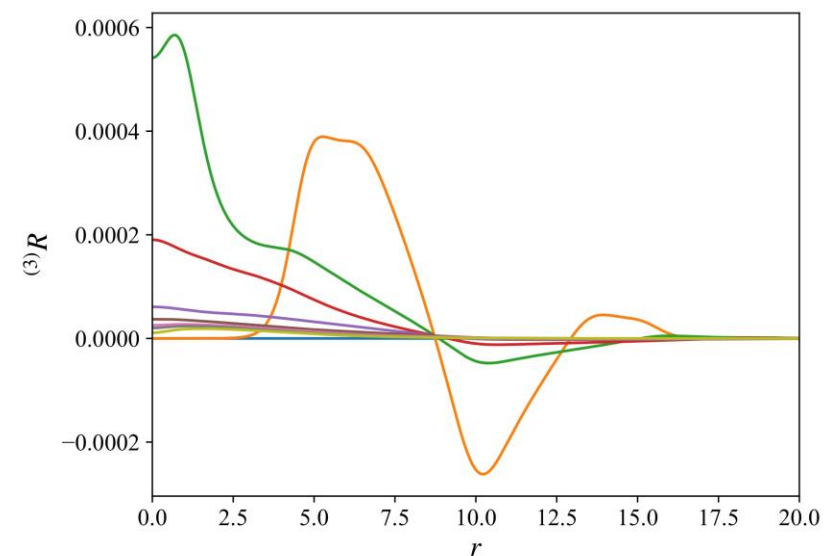
small density contrast

Outcome 3: No PBH Formation (Dispersal)

- $V_F = 5 \times 10^{-7} H_i^2$, no PBH formed
- Energy is radiated away as outgoing scalar waves



no trapped surface



relax back toward zero

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Defining the criteria

- Horizon-based diagnostics are equivalent to the Schwarzschild condition

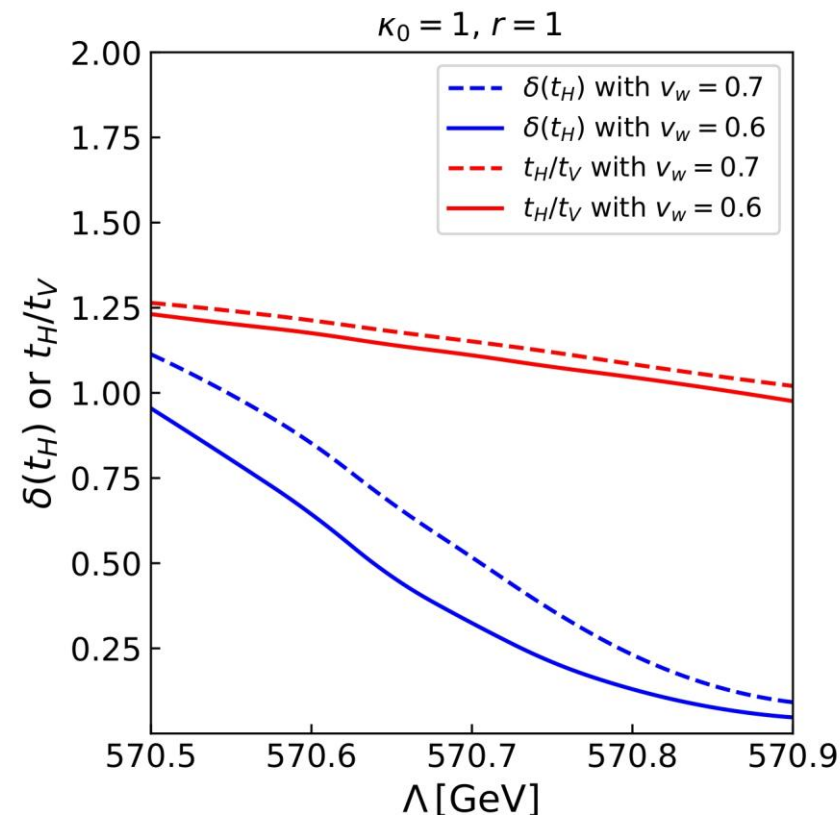
$$\Theta^+ \Theta^- = \frac{2}{R^2} (U^2 - \Gamma^2) = 0 \implies \frac{2M}{R} = 1$$

1. Time scale ratio t_H/t_V

- Horizon crossing time $t_H = \frac{r_i^2}{4t_i}$
- Vacuum energy domination $t_V = \frac{1}{2} \sqrt{\frac{3}{8\pi\rho_V}}$
- $t_H/t_V \gtrsim 1$: type B PBH
- $t_H/t_V \lesssim 1$: type A PBH or dispersion

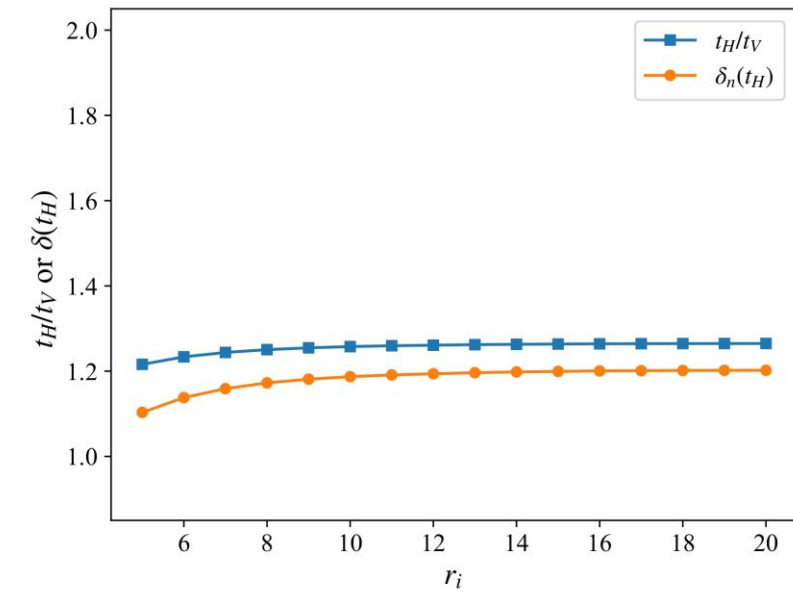
2. Local density contrast $\delta(t_H)$

$$\delta(t_H) = \frac{\rho_{\text{in}}(t_H) - \rho_{\text{out}}(t_H)}{\rho_{\text{out}}(t_H)}$$

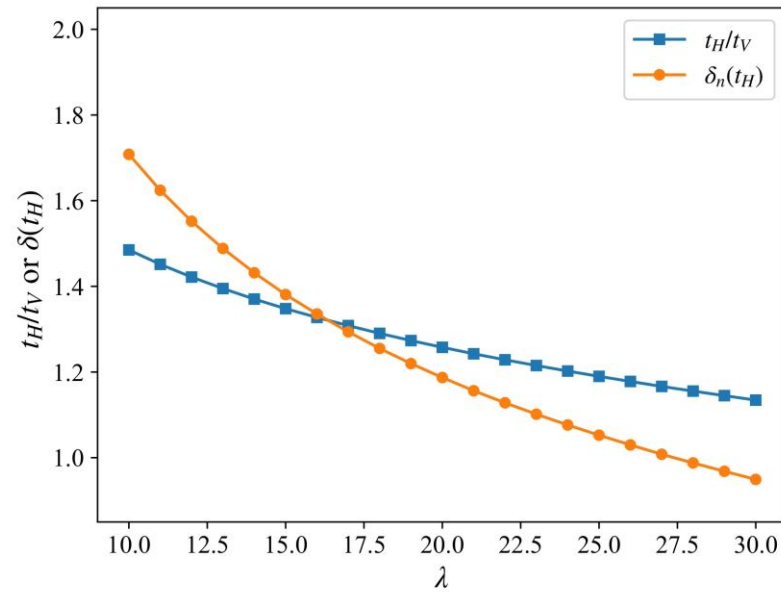


Threshold for type B PBH

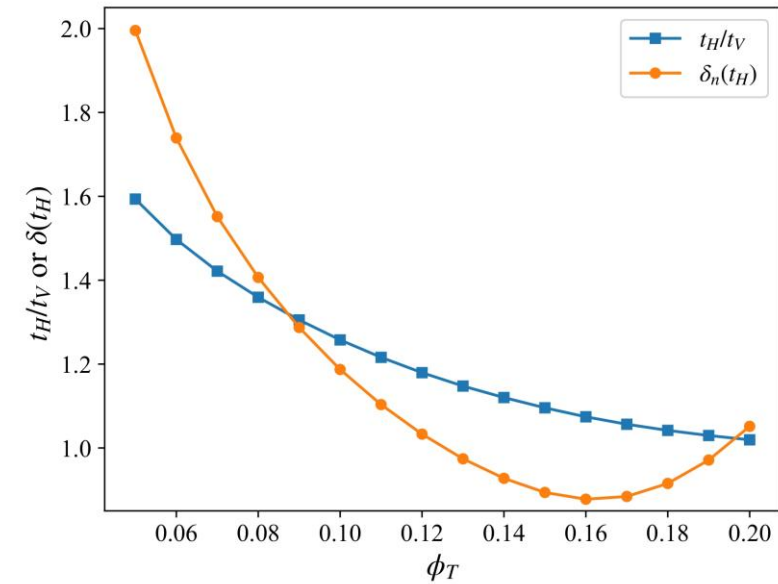
- Use a bisection search to find critical thresholds



insensitive to the initial FVD size



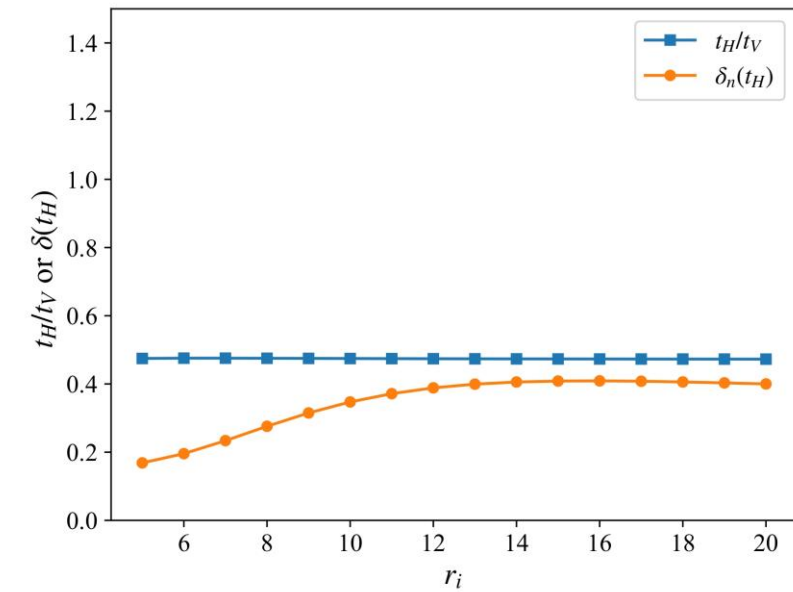
change in t_H/t_V is relatively small



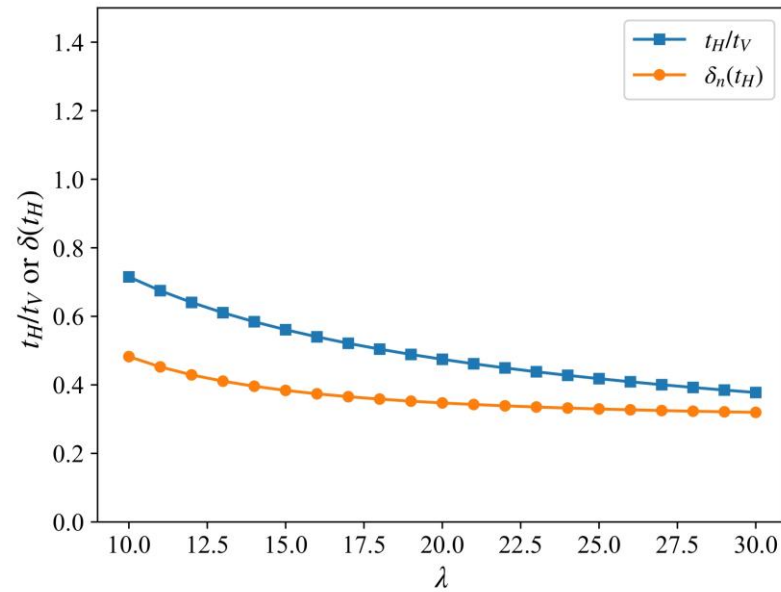
$\delta(t_H)$ is unstable

Threshold for type A PBH

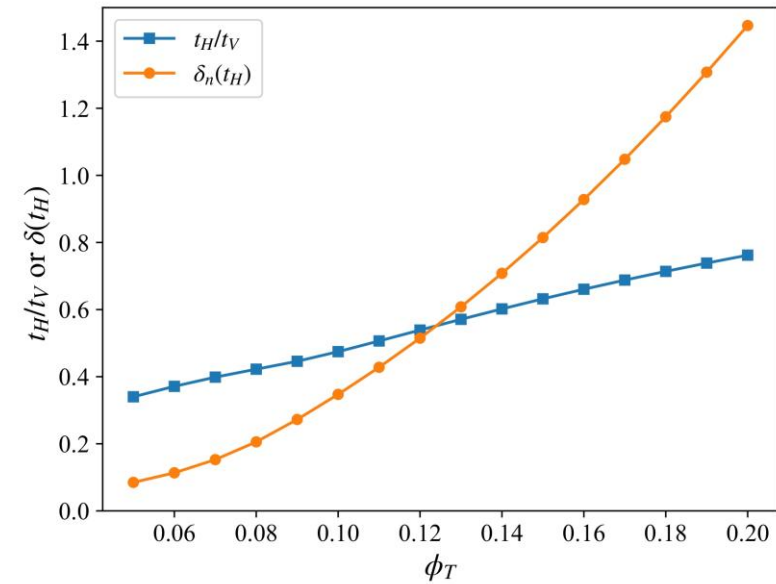
- Use a bisection search to find critical thresholds



For small r_i the thin-wall configuration is broken before horizon crossing



$\delta(t_H) \sim 0.35 - 0.5$



$\delta(t_H)$ is unstable

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Conclusions and discussions

- **Dynamics:** Identified three outcomes:
 - Type B: Supercritical PBH with baby universe
 - Type A: Subcritical PBH from direct wall collapse
 - Dispersal: No PBH formed

- **PBH formation criteria:**
 - Validated t_H/t_V as a more robust criterion than $\delta(t_H)$

Threshold	t_H/t_V	$\delta(t_H)$
Type B	$\sim 1.1 - 1.6$	$\sim 1.0 - 1.7$ (unstable)
Type A	$\sim 0.35 - 0.7$	$\sim 0.35 - 0.5$ (unstable)

- **Future work:**
 - Non-spherical simulations (3D)
 - Finite-temperature effects and scalar-fluid couplings
 - Calculation of PBH abundance based on t_H/t_V statistics

Thank you!